Fundamentals of Machine Learning [CSS 3131]

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- What is ML?

  ML is a set of algorithms that can learn to perform tasks such as prediction and classification effectively using data.

> A computer program is said to learn from experience E with respect to some class of tasks T, and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.
>
> - Tom Mitchell (1997)

> A field of study that gives computers the ability to learn without being explicitly programmed.
>
> - Arthur Samuel (1959).

Need for Machine Learning?

- High volume of available Data
- Cost of storage has reduced
- Unavailability of complex algorithms

Important Objectives of Machine Learning:

- To process archived data for organizations
- To make better decisions
- To design new products and
- To improve business processes to develop decision support systems

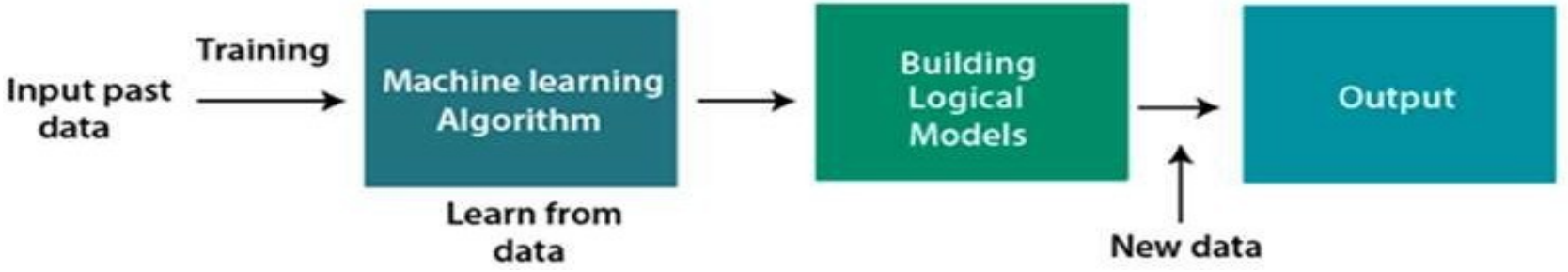
Introduction to Machine Learning

- In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions.
- But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.
- Machine Learning is said as a subset of artificial intelligence that is mainly concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own.



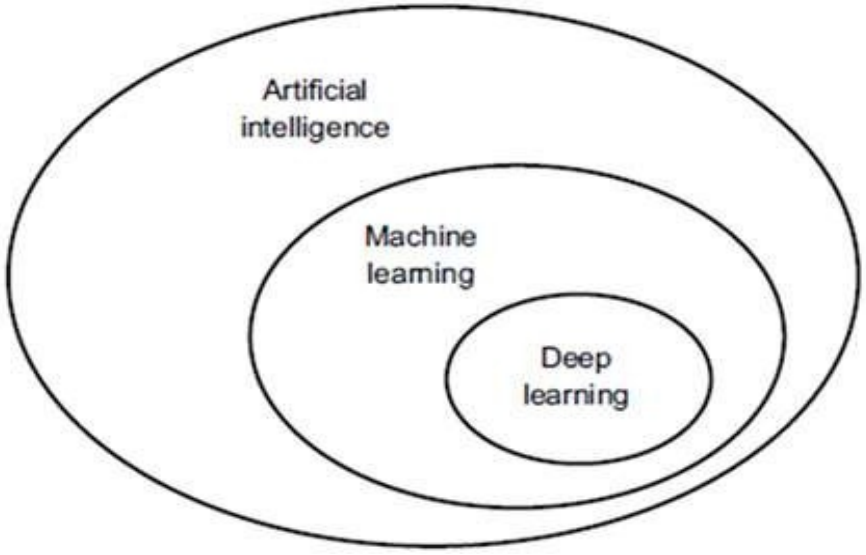
Introduction to Machine Learning

- A Machine Learning system learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it.
- The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.
- Suppose we have a complex problem, where we need to perform some predictions. Instead of writing a code for it, we need to feed the data to generic algorithms, and with the help of these algorithms, machine builds the logic as per the data and predict the output.



What is Artificial Intelligence (AI)?

- Concisely, **AI can be described** as the effort to **automate Intellectual tasks** normally performed by humans.
- AI is a general field that encompasses machine learning and deep learning
- AI also includes many more approaches that may not involve any learning.



Artificial intelligence

- Symbolic AI: Dominant paradigm in AI from the 1950s -1980
- Reached peak popularity during the expert systems boom of the 1980s
- Early chess programs only involved **hardcoded rules crafted by programmers** and did not qualify as machine learning.
- For a fairly long time, most experts believed that human-level artificial intelligence could be achieved by having programmers handcraft a sufficiently large set of explicit rules for manipulating knowledge stored in explicit databases.
- This approach is known as symbolic AI.

Note: The origins of chess are debated, but most historians agree that the game's ancestor, *chaturanga*, originated in India around the 6th century AD.

Artificial intelligence

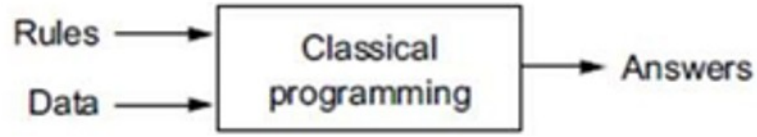
- Symbolic AI to Machine Learning
- Symbolic AI proved suitable to solve well-defined, logical problem
  - such as playing chess
- Turned out to be intractable to figure out explicit rules for solving more complex, fuzzy problem, image classification, speech recognition, or natural language translation.
- A new approach arose to take symbolic AI's place: machine learning

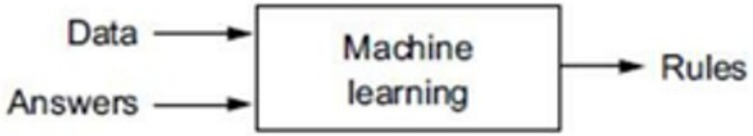
Machine Learning

- Machine learning looks at the input data and the corresponding answers, and **figures out what the rules should be**
- A machine learning **system is trained rather than explicitly programmed.**
- It is presented with many examples relevant to a task, and it finds statistical structure in these examples that eventually allows the **system to come up with rules for automating the task.**

Machine Learning: a new programming paradigm

Deep Learning with Python – Francois Chollet

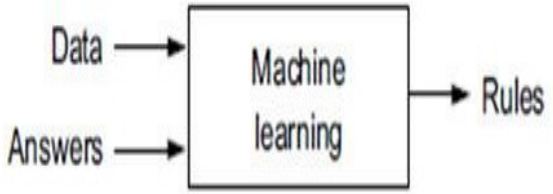




Machine Learning

- Unlike statistics, machine learning tends to deal with large, complex datasets (such as a dataset of millions of images, each consisting of tens of thousands of pixels)
- Classical statistical analysis such as Bayesian analysis would be impractical.
- Machine learning, and especially deep learning, exhibits comparatively little mathematical theory and is fundamentally an engineering discipline

Machine Learning



To do machine learning, we need three things:

1.Input data points—

- For instance, if the task is speech recognition, these data points could be sound files of people speaking.
- If the task is image tagging, they could be pictures.

2.Examples of the expected output—

- In a speech-recognition task, these could be human-generated transcripts of sound files.
- In an image task, expected outputs could be tags such as “dog,” “cat,” and so on.

3. **A way to measure whether the algorithm is doing a good job—**

- This is necessary to determine the **distance between the algorithm’s current output and its expected output.**
- The measurement is used as a feedback signal to adjust the way the algorithm works.
- This adjustment step is what we call learning.

Motivation

- Over the past two decades, Machine Learning has become one of the mainstays of information technology
- With the ever-increasing amounts of data becoming available, there is good reason to believe that **smart data analysis will become more pervasive** as a necessary ingredient for technological progress

CSS 3131 Fundamentals of Machine Learning [3 0 3 4]

Machine Learning Basics: Types of Machine Learning, Supervised vs. Unsupervised Learning, Parametric vs. non-parametric models., Instance Based learning – k-nearest neighbors, Simple Regression Models: Linear, Logistic, Cost functions, Gradient Descent, Batch Gradient Descent, Overfitting, Model Selection, No free lunch theorem, bias/variance trade-off, union and Chernoff bounds, VC dimensions. Bayesian Models: Bayesian concept learning, Bayesian Decision Theory, Naïve Bayesian, Laplacian Correction, Bayesian Belief Networks. Tree Models: information theory, decision tree induction, tuning tree size, ID3,C4.5, CHAID, Decision Stump. Support Vector Machines: kernel functions, Regression Models: Ridge and Lasso Regression, GLM and the exponential Family. Bagging algorithm, Random Forests, Grid search and randomized grid search, Partial dependence plots. Ensembling and Boosting Algorithms: Concept of weak learners, Adaptive Boosting, Extreme Gradient Boosting (XGBoost). Artificial Neural Networks: Perceptron, Back propagation, Hopfield Network. Curse of Dimensionality: Factor Analysis, Principal Component Analysis (PCA), Difference between PCAs and Latent Factors

References

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Spam filter - ML Application

- Spam filter is a Machine Learning program
- Given examples of spam emails (e.g., flagged by users) and examples of regular (nonspam, also called “ham”) emails can learn to flag spam.
- The examples that the system uses to learn are called the training set.
- Each training example is called a training instance (or sample).
- Task T is to flag spam for new emails
- Experience E is the training data
- Performance measure P needs to be defined;
  - We can use the ratio of correctly classified emails.
  - This particular performance measure is called accuracy, and it is often used in classification tasks.

Spam filter Application using traditional programming technique

- What spam typically looks like?

1. Notice that some words or phrases (such as “4U,” “credit card,” “free,” and “amazing”) tend to come up a lot in the subject line.

- Perhaps a few other patterns in the sender’s name, the email’s body, and other parts of the email could also be noticed

2. Write a detection algorithm for each of the patterns noticed, and program would flag emails as spam if a number of these patterns were detected.

- 3. Test our program and repeat steps 1 and 2 until it was good enough

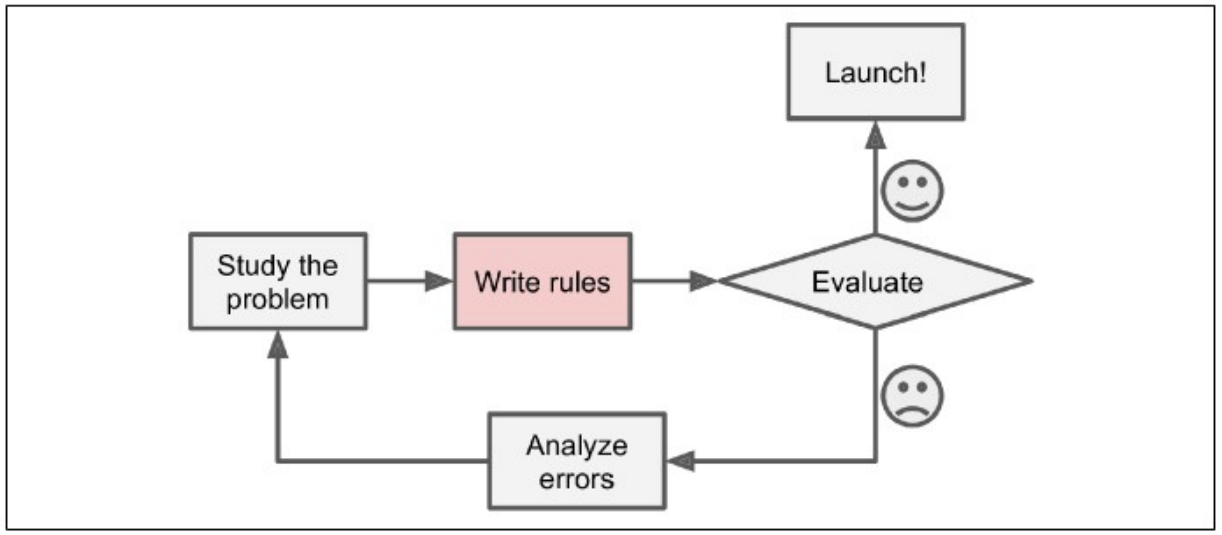
to launch.

Problem?

Program will likely become a long list of complex rules—hard to maintain

Spam filter Application using traditional programming technique

- Write a spam filter using traditional programming techniques

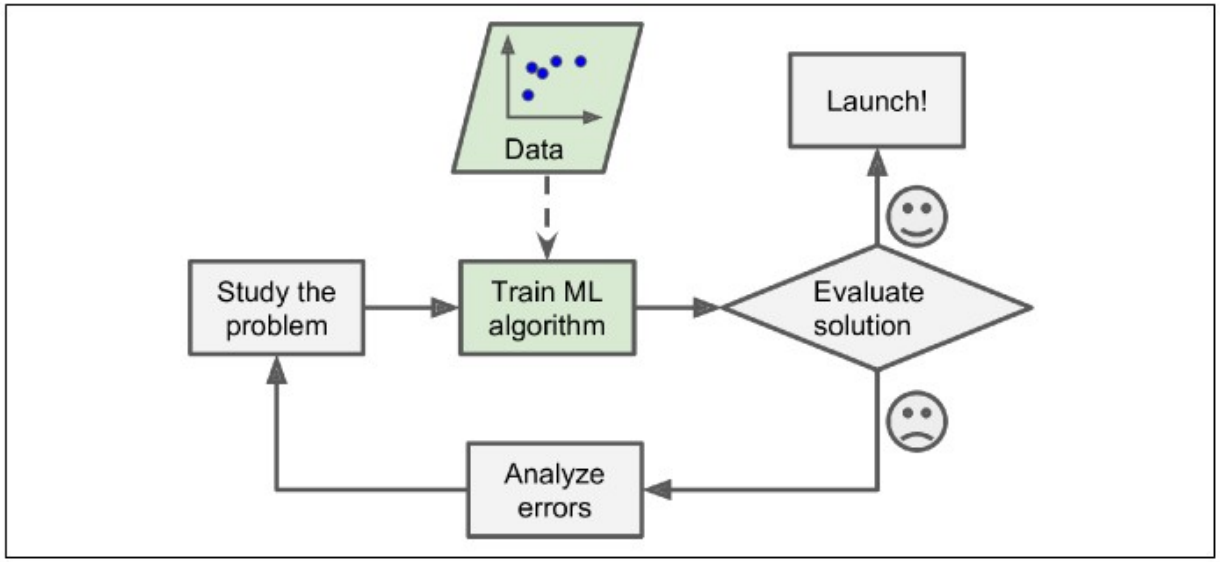


Spam filter using ML Approach

- a spam filter based on Machine Learning techniques automatically learns which words and phrases are good predictors of spam
- by detecting unusually frequent patterns of words in the spam examples compared to the ham examples
- The program is much shorter, easier to maintain, and most likely more accurate.

Spam filter using ML Approach

- Write a spam filter using ML Approach



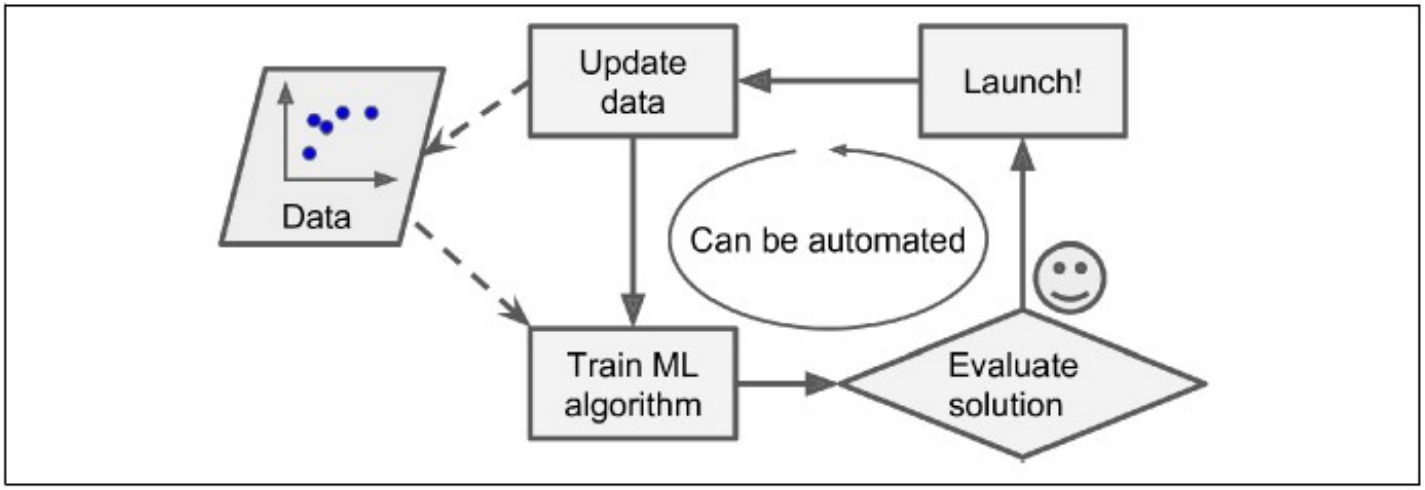
Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow Aurélien Géron

Spam filter using ML Approach

- Automatically adapting to change
- If spammers keep working around our spam filter, we need to keep writing new rules forever
- Ex: What if spammers notice that all their emails containing “4U” are blocked?
- They might start writing “For U” instead.
- A spam filter using traditional programming techniques would need to be updated to flag “For U” emails.
- In contrast, a spam filter based on Machine Learning techniques automatically notices that “For U” has become unusually frequent in spam flagged by users, and it starts flagging them without our intervention

Spam filter using ML Approach

- Write a spam filter using ML Approach- Automatically adapting to change



Other Applications using ML Approach

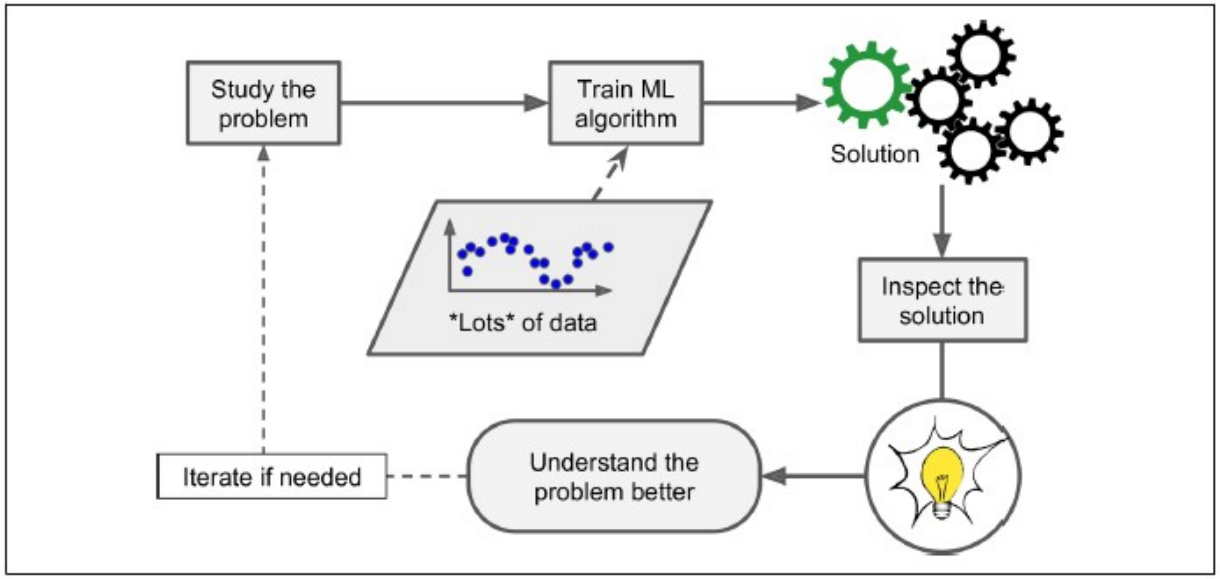
- Another area where Machine Learning shines is for problems that either
  - are too complex for traditional approaches or
  - have no known algorithm.
- Ex: Speech recognition
- To write a program capable of distinguishing the words “one” and “two.”
- We might notice that the word “two” starts with a high-pitch sound (“T”)
- so we could hardcode an algorithm that measures high-pitch sound intensity and use that to distinguish ones and twos
- Problem?
- But this technique will not scale to thousands of words spoken by millions of very different people in noisy environments and in dozens of languages.
- The best solution is to write an algorithm that learns by itself, given many example recordings for each word.

Other Applications using ML Approach

- Machine Learning can help humans learn
- ML algorithms can be inspected to see what they have learned (although for some algorithms this can be tricky).
- Ex: once a spam filter has been trained on enough spam, it can easily be inspected to reveal
  - the list of words and
  - combinations of words that it believes are the best predictors of spam.
- Sometimes this will reveal unsuspected correlations or new trends, and thereby lead to a better understanding of the problem.
- Applying ML techniques to dig into large amounts of data can help discover patterns that were not immediately apparent.
- This is called data mining

Other Applications using ML Approach

Machine Learning can help humans learn



Why Use Machine Learning?- Summary

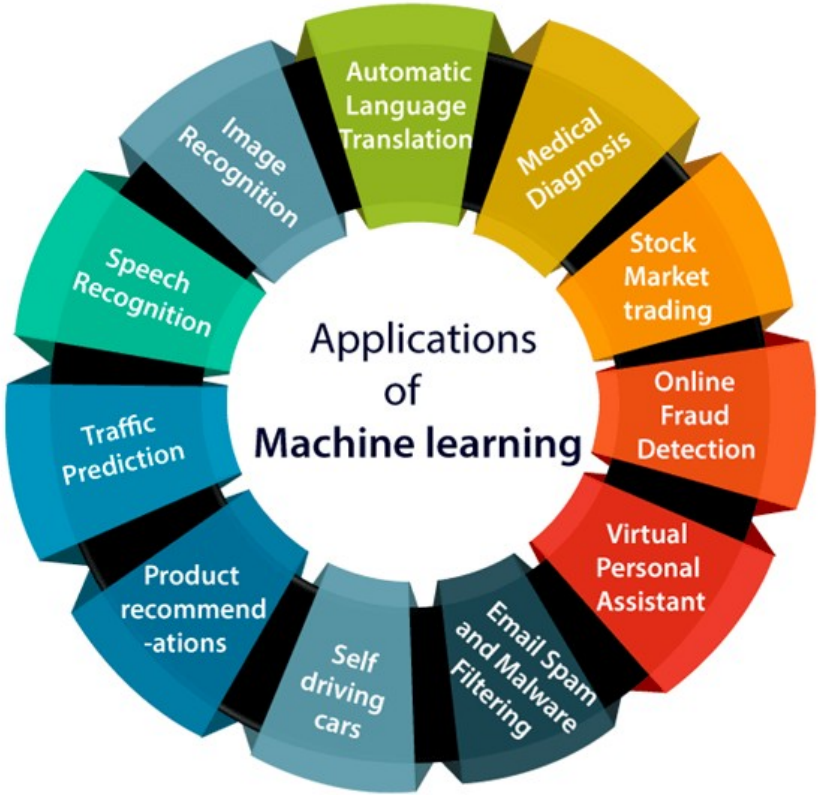
Machine Learning is great for:

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules:
  - one Machine Learning algorithm can often simplify code and perform better than the traditional approach.
- Complex problems for which using a traditional approach yields no good solution:
  - the best Machine Learning techniques can perhaps find a solution.
- Fluctuating environments: a Machine Learning system can adapt to new data.
- Getting insights about complex problems and large amounts of data.

EXAMPLES OF APPLICATIONS IN DIVERSE FIELDS

- Machine learning is a growing technology used to mine knowledge from data (popularly known as data mining field).
- Wherever data exists, things can be learned from it.
  - Whenever, there is excess of data, the mechanics of learning must be automatic.
  - Machine learning technology is meant for automatic learning from voluminous datasets.
- Applications emerge not from machine learning experts, nor from the data itself, but from people who work with the data and the problems from which it arises.

EXAMPLES OF APPLICATIONS IN DIVERSE FIELDS



Classification of machine learning algorithms

- Supervised Learning
  - Classification
  - Regression
- Unsupervised Learning
  - Clustering
  - Association
- Semi-supervised Learning
- Reinforcement Learning

Supervised learning

- Uses labelled data (the training data fed to the algorithm are labelled)
- Here, the training data you feed to the algorithm includes the **desired solution**, called **labels**
- Two main categories**: Classification and Regression**
- ML has two important stages: Training and Testing
- In the Training stage (first stage), the algorithm takes a labelled dataset(input) and starts learning
- After training, the samples are processed, and the model is generated
- In the second stage (Testing) , the constructed model is tested with the test samples

Classification Model

- In classification, input attributes are independent variables, and target attributes are dependent variables
- Here the **output will be discrete value** [**Output:** Discrete, separate groups (e.g., yes/no, cat/dog, spam/not spam)]
- Relationship between dependent (target) and independent(input) is represented in the form of structure called classification
- Important Classification algorithms in ML: Logistic Regression, Support Vector Machine, KNN, Naïve Bayes, etc

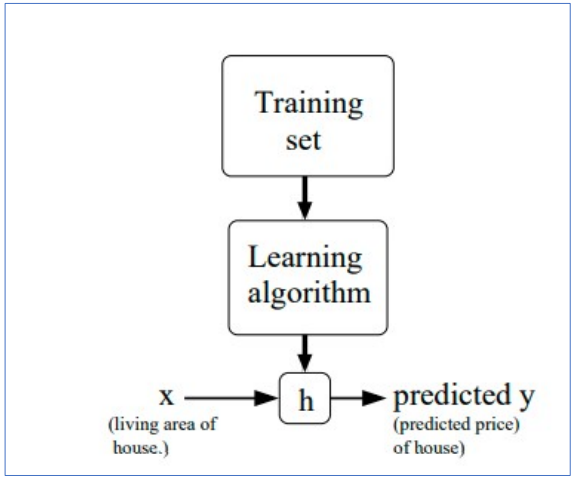
Regression Model

- Unlike classification (wherein the output will be a discrete value, either 0/1, setosa/virginica), in regression, the **output will be a continuous value**

  [**Output:** Continuous real numbers within a range (e.g., price, temperature, weight]
- Important Regression algorithms in ML: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, etc

Supervised learning

- predict y from x
- Given a labelled set of input-output pairs, Map input x to output y
- Given: Training set $\{(x_i, y_i) \mid i = 1 \ldots n\}$
- Find: A good approximation to
  - $f : X \rightarrow Y$ where $y \in \{1, \ldots, C\}$,
- Classification – y is categorical
- Regression – y is a real value
- Formalize the problem is as function approximation.
- $y = f(x)$ for some unknown function f
  - Goal of learning is to estimate the function f
  - Given a labeled training set
  - then to make predictions using $\hat{y} = \hat{f}(x)$.
- Generalization: to make predictions on novel inputs
- Examples - spam detection, digit recognition, stock prices



Learning with Teacher (Supervised Learning)

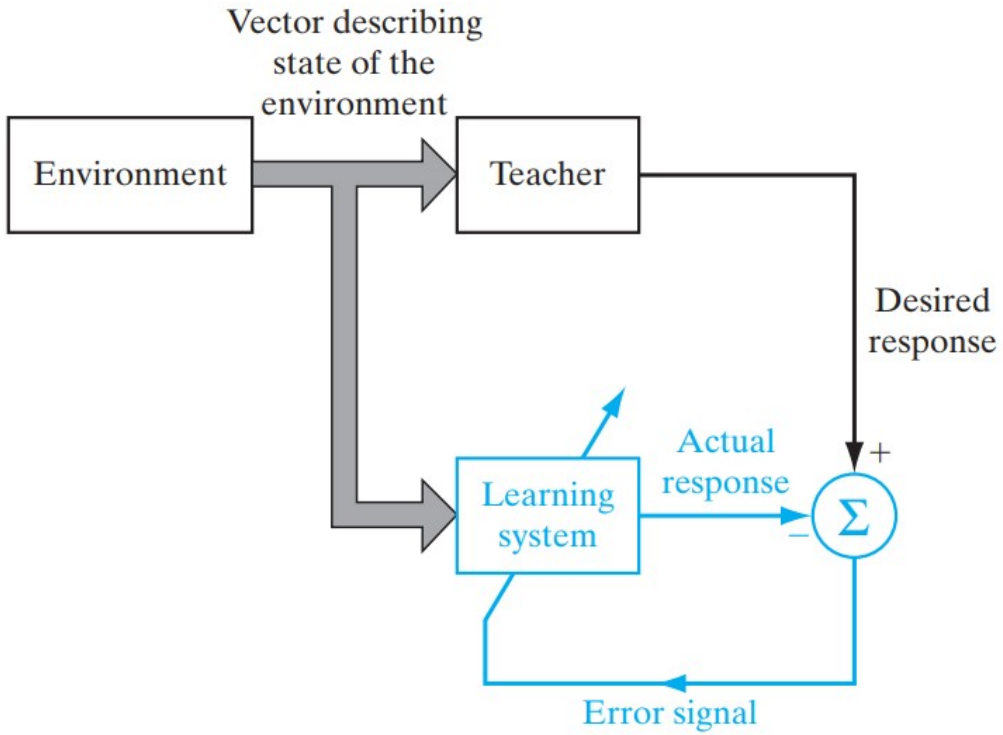
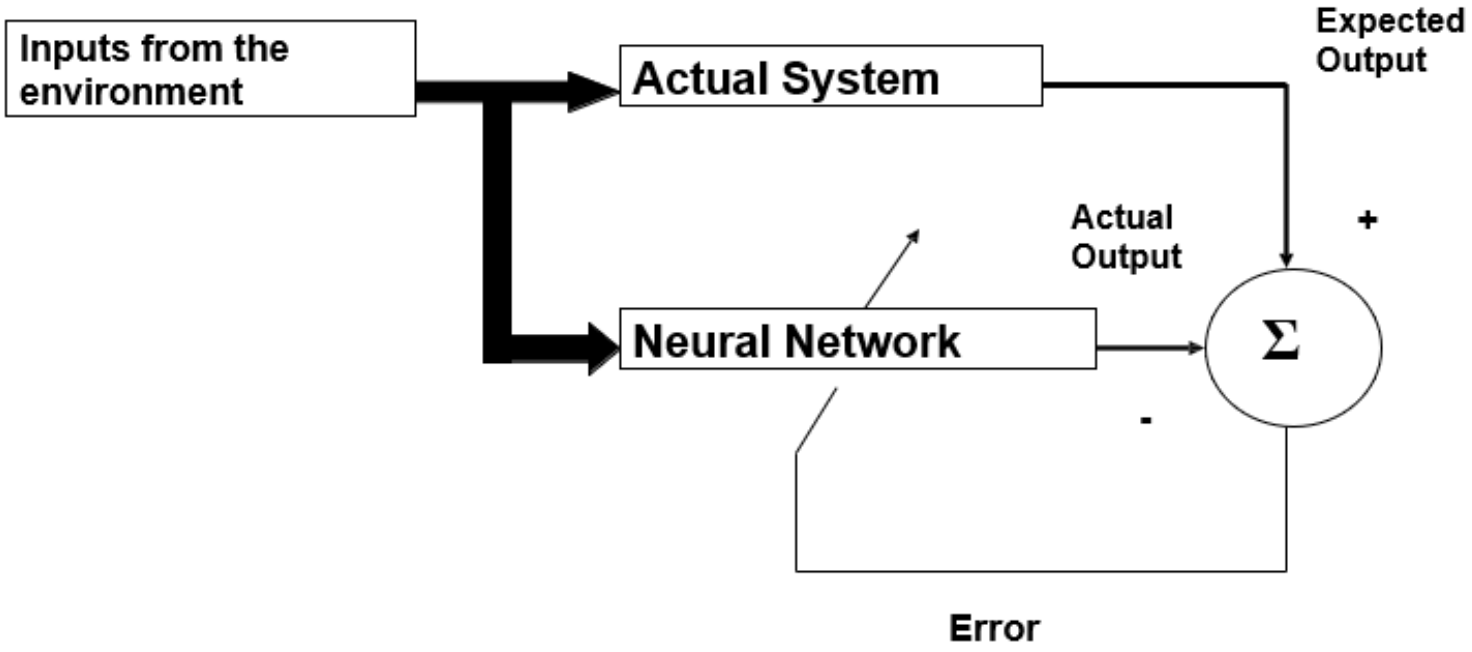


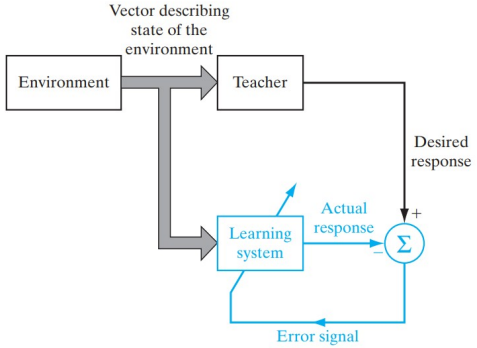
FIGURE 24 Block diagram of learning with a teacher; the part of the figure printed in red constitutes a feedback loop.

Learning with Teacher (Supervised Learning)

- Child learns from a teacher
- Each input vector requires a corresponding target vector.
- Training pair=[input vector, target vector]



Learning with Teacher (Supervised Learning)

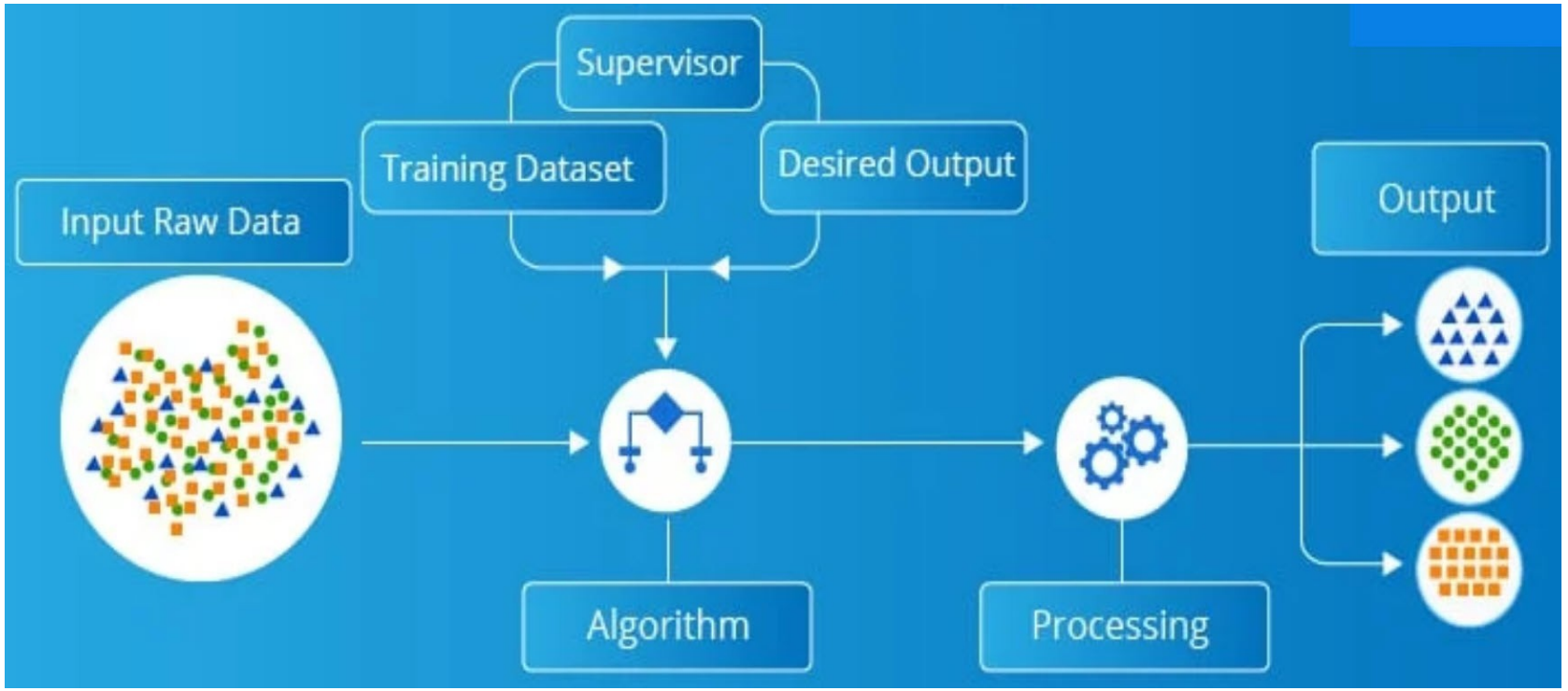


- ✓ The supervised learning strategy consists of having available the **desired outputs** for a given set of **input signals**; in other words, each **training sample** is composed of the *input signals and their corresponding outputs*.
- ✓ Henceforth, it requires a table with **input/output data**, also called an attribute/value table, which represents the process and its behavior.
- ✓ It is from this information that the neural structures will formulate a “hypothesis” about the system being learned.
- ✓ Donald Hebb proposed the **first supervised learning strategy** in 1949, inspired by neurophysiological observations (Hebb 1949)

Learning with Teacher (Supervised Learning)

- The network is trained by providing it with input and matching output patterns.
- These input-output pairs are usually provided by an external teacher.
- The network then processes the inputs and compares its resulting outputs against the **desired outputs**.
- Errors are then propagated back through the system, causing the system to adjust the weights which control the network.
- This process occurs over and over as the weights are continually tweaked. The set of data which enables the training is called the "**training set**".
- During the training of a network, the same set of data is processed many times as the connection weights are ever refined.
- **Architectures (models):** Perceptron, Feed-forward, Radial Basis Function, Support Vector Machine.

Learning with Teacher (Supervised Learning)



Learning without Teacher (Unsupervised Learning)

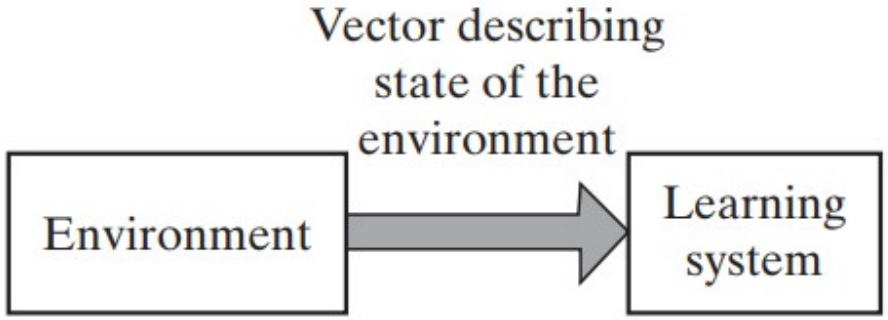


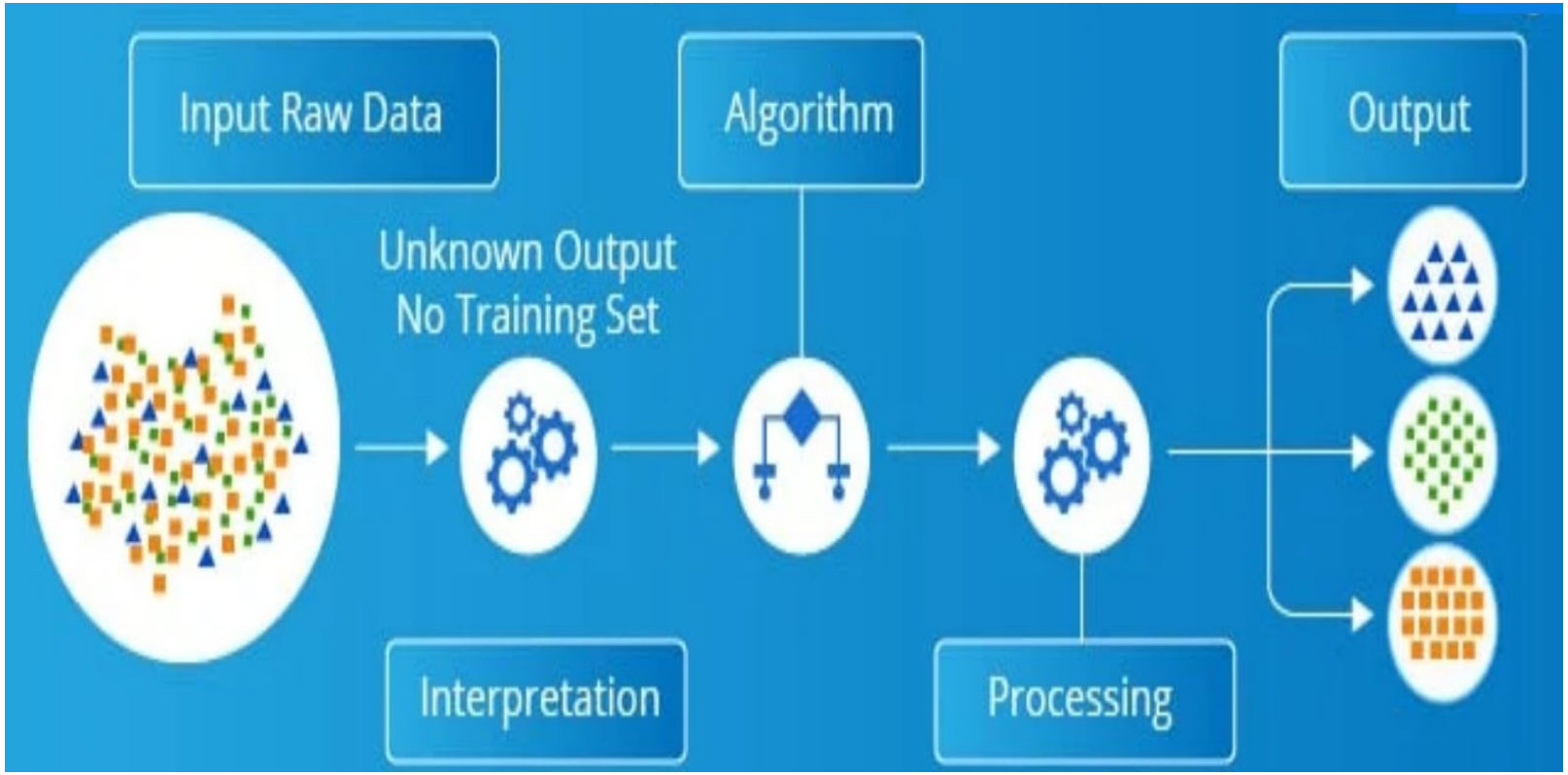
FIGURE 26 Block diagram of unsupervised learning.

Learning without Teacher (Unsupervised Learning)

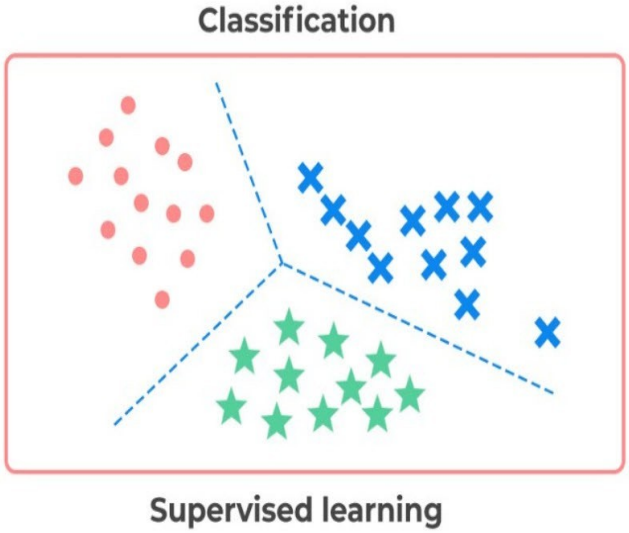
- Self-organized learning, no external teacher or critic to oversee the learning process.
- The system is supposed to discover statistically salient features of the input population.
- Unlike supervised learning, there is no a priori set of categories into which the patterns are to be classified; rather, the system must develop its own representation of the input stimuli.
- How a fish or tadpole learns
- All similar input patterns are grouped together as clusters.
- If a matching input pattern is not found, a new cluster is formed.

- **Architecture (models):** Self-organizing maps, Hopfield networks.

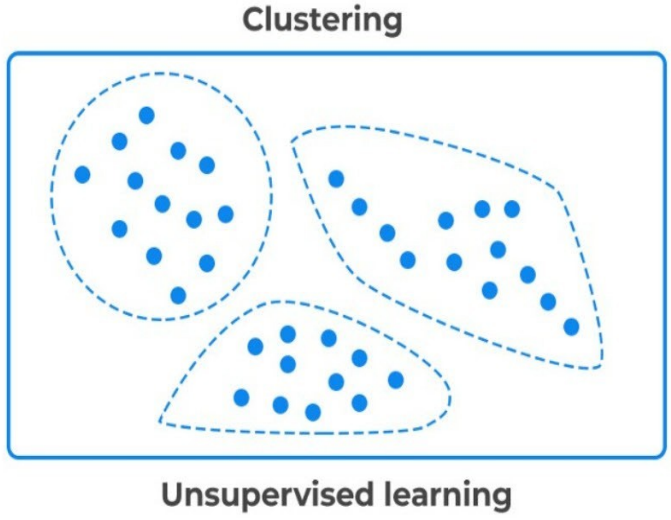
Learning without Teacher (Unsupervised Learning)



Supervised *Vs* Unsupervised Learning



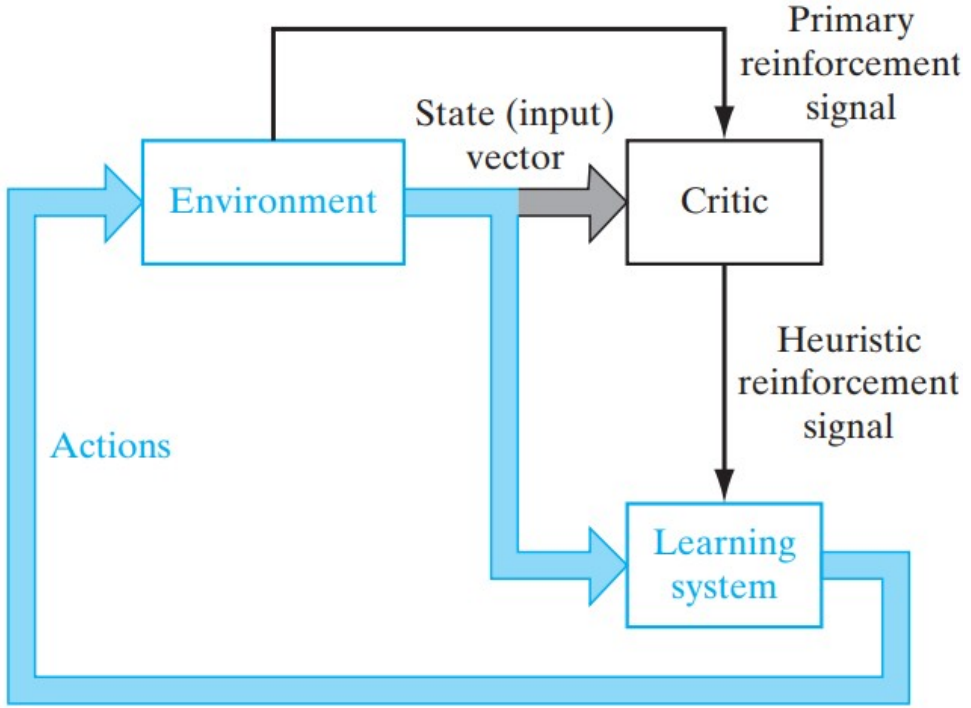
Supervised learning



Unsupervised learning

Learning without Teacher (Reinforcement Learning)

FIGURE 25 Block diagram of reinforcement learning; the learning system and the environment are both inside the feedback loop.



Learning without Teacher (Reinforcement Learning)

- Learning of an input–output mapping is performed through continued interaction with the environment in order to minimize a scalar index of performance.
- Computer agent learns to perform a task through **repeated trial-and-error interactions** within a dynamic environment.
- Neural networks can **approximate value functions**, policies, or both, enabling RL to handle high-dimensional state and action spaces.

When Reinforcement learning is used?

- If **less information is available** about the target output values (critic information).
- Learning based on this critic information is called reinforcement learning, and the feedback sent is called reinforcement signal.
- Feedback in this case is only evaluative and not instructive.

Learning without Teacher (Reinforcement Learning)



Supervised *Vs* Unsupervised *Vs* Reinforcement Learning

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	Learns from labeled data	Learns from unlabeled data	Learns by interacting with an environment
Data Type	Labeled data (input-output pairs)	Unlabeled data (only inputs)	Sequential data with rewards or penalties
Objective	Predict output (labels) for new data	Find hidden patterns or structures	Maximize cumulative reward over time
Examples of Tasks	Classification, Regression	Clustering, Dimensionality Reduction	Game playing, Robotics, Self-driving cars
Feedback Type	Direct feedback (correct answer provided)	No feedback	Delayed feedback via rewards or punishments
Learning Process	Learns a mapping from inputs to outputs	Learns data distribution or groupings	Learns policy or value function from experience
Algorithms	Linear Regression, SVM, Decision Trees, Neural Nets	K-Means, PCA, Autoencoders, Hierarchical Clustering	Q-Learning, SARSA, Deep Q Networks (DQN), Policy Gradient
Output	Predictive model	Data structure or transformed features	Optimal strategy or policy
Real-world Examples	Spam detection, Medical diagnosis, Stock prediction	Customer segmentation, Anomaly detection	AlphaGo, Recommendation systems, Autonomous robots

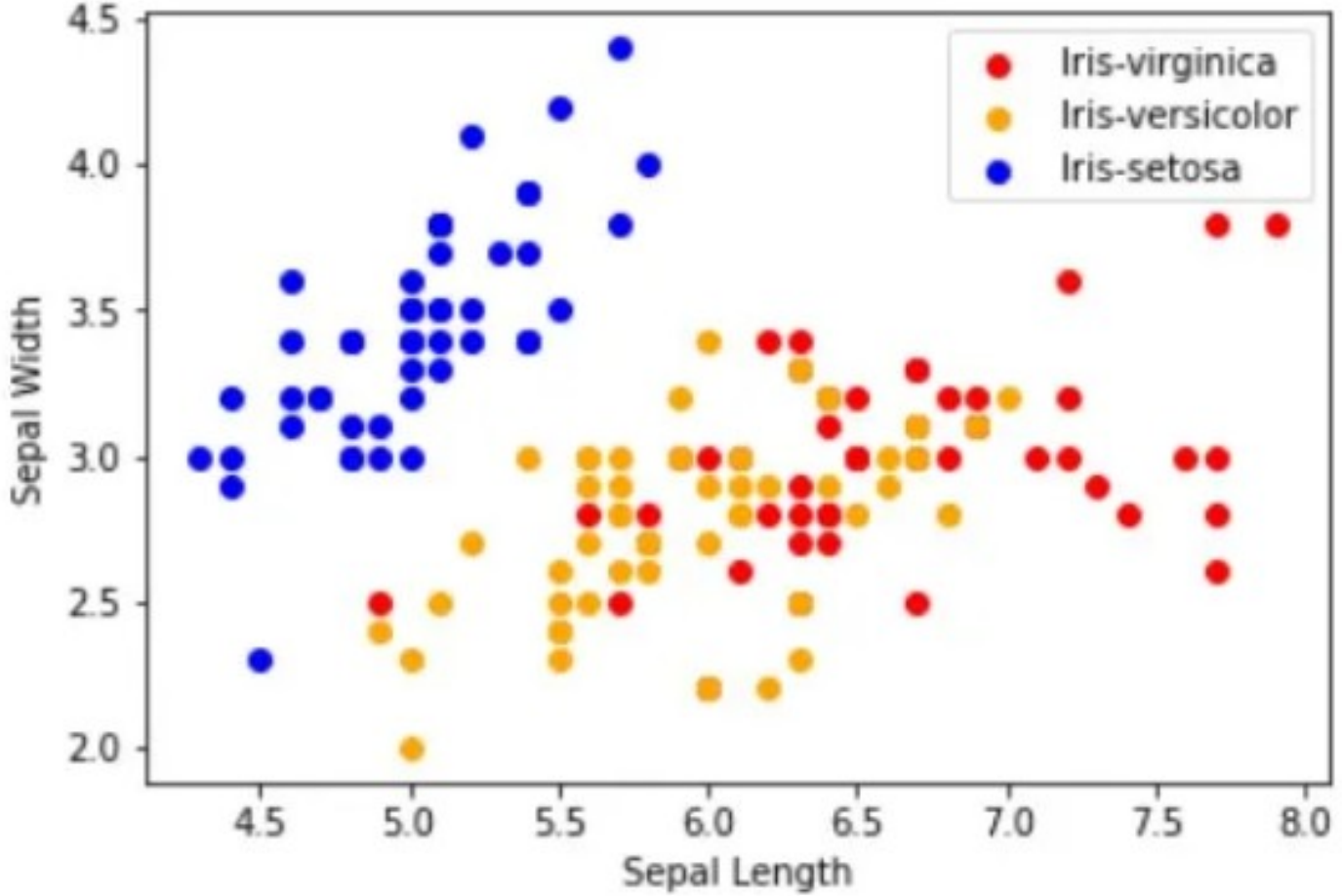
Example: Iris Data Set

Suppose we have to classify the flowers into 3 categories from the Iris dataset based on the attributes Petal length, Petal width, sepal length & sepal width.
The sample dataset is given below:

Sepal Length (cm)	Sepal Width (cm)	Petal Length (cm)	Petal Width (cm)	Species (Category)
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3.0	1.4	0.2	Iris-setosa
4.7	3.2	1.3	0.2	Iris-setosa
7.0	3.2	4.7	1.4	Iris-versicolor
6.4	3.2	4.5	1.5	Iris-versicolor
6.9	3.1	4.9	1.5	Iris-versicolor
6.3	3.3	6.0	2.5	Iris-virginica

Example: Iris Data Set

Consider the scatter plot with sepal length (x-axis) & sepal width(y-axis) for the given Iris Dataset for 150 records (50 each for 3 types of flowers)



Example: Iris Data Set

Suppose that we have to classify the flowers among 3 different categories from the Iris dataset based on the attributes sepal length & sepal width

Sepal Length (cm)	Sepal Width (cm)	Flower Category
5.1	3.5	Iris setosa
4.9	3.0	Iris setosa
7.0	3.2	Iris versicolor
6.4	3.2	Iris versicolor
6.3	3.3	Iris virginica
5.8	2.7	Iris virginica

Example: Iris Data Set



(a) (b) (c)

Figure 1.3 Three types of iris flowers: setosa, versicolor and virginica. Source: `http://www.statlab.uni-heidelberg.de/data/iris/` . Used with kind permission of Dennis Kramb and SIGNA.

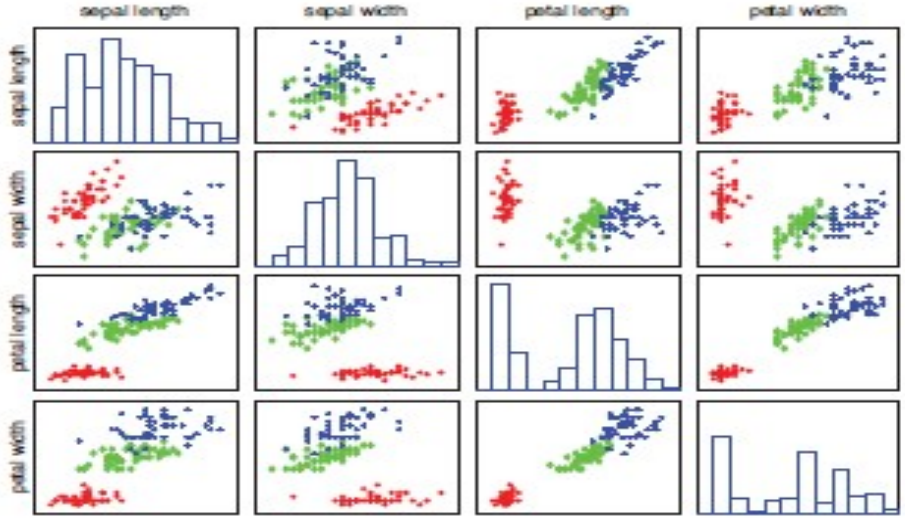


Figure 1.4 Visualization of the Iris data as a pairwise scatter plot. The diagonal plots the marginal histograms of the 4 features. The off diagonals contain scatterplots of all possible pairs of features. Red circle = setosa, green diamond = versicolor, blue star = virginica. Figure generated by `fisheririsDemo`.

Linear Regression

Linear Regression is a simple and powerful model approach for predicting a

- numeric response from a set of one or more independent variables.

It models the relationship between a dependent variable y and one or

- more independent variables X.

In Linear regression, the **goal is to find a line (or hyperplane) that best fits**

- **the data points by minimizing the error** between predicted and actual values.

Dependent and Independent Variable:

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y -> variable we wish to predict, i.e. dependent, Response, or Target variable

- X -> variable used to predict Y, independent, predictor variable

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Linear Regression

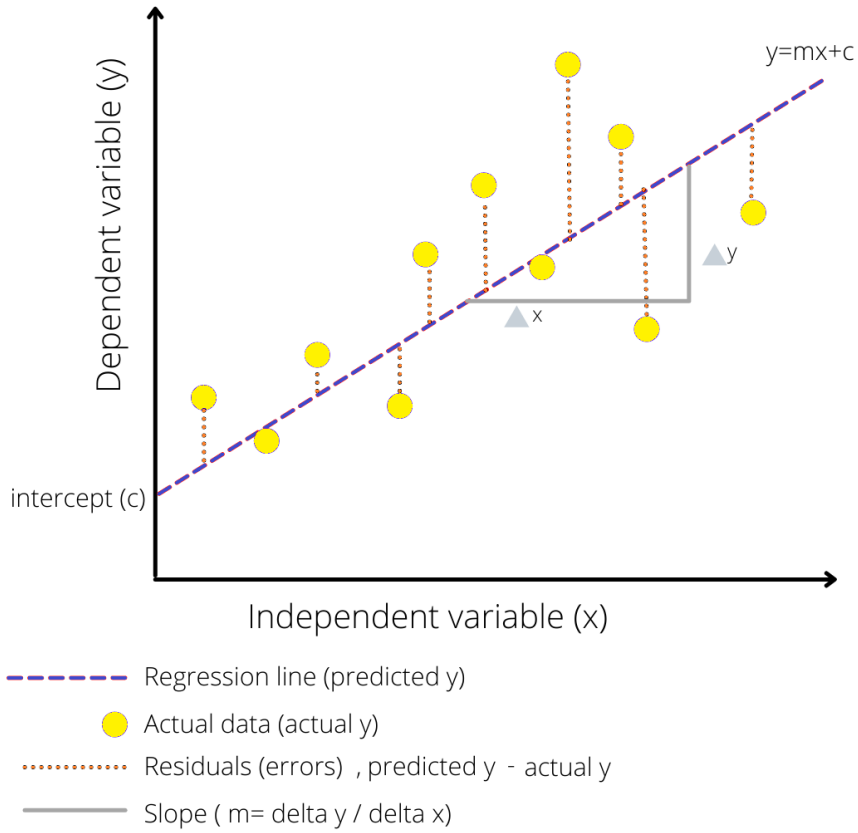
Based on the type of input data this algorithms can be applied in three

- different forms:

  Simple Linear Regression

- Multi Linear Regression
- Polynomial Regression

Regression Model



Breaking Down the Equation
(y): The target or output value you want to predict.
(x): The input feature or data point you use to make the prediction.
(m): The slope of the line, showing how much y changes when (x) goes up by one unit.
(c): The y-intercept, showing the value of (y) when (x) is equal to zero

Simple Linear Regression

- The simple linear regression model relies on:

$$y = mx + c$$

to predict a dependent output y from an independent input x using a straight line

- where:

m= slope (weight)

Simple Linear Regression

To calculate the value of feature weights or slope (**m**) and intercept (**c**) :

$$m = \frac{\sum_{i=1}^{m}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{m}(x_i - \bar{x})^2}$$

$$c = \bar{y} - mx$$

Simple Linear Regression

- Equation of MSE for linear regression model :-

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- For every true value y and predicted value $\hat{y}$ for the same data, we calculate the error.

- **RMSE (Root Mean Squared Error)**: Takes the square root of the MSE value

RMSE = sqrt(MSE)

Simple Linear Regression

To calculate the **coefficient of determination, or R^2**, in linear regression, you **divide the Residual Sum of Squares (SS_{res}) by the Total Sum of Squares (SS_{tot}) and subtract the result from** 1. Stack Exchange

The core formula is:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

1. Find the mean of your observed data

Calculate the average of all actual dependent variable values (y).

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

- y_i: The actual observed values.
- n: The total number of data points. GeeksforGeeks

Simple Linear Regression

2. Compute the Total Sum of Squares (SS_{tot})

This measures the total variation in your observed data from their mean value.

$$SS_{\text{tot}} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

3. Compute the Residual Sum of Squares (SS_{res})

Also known as the Sum of Squared Errors (SSE), this measures the variation that your regression line fails to explain. First, determine your predicted values ($\hat{y}_i$) using your linear regression equation ($\hat{y} = mx + b$). Then calculate:

$$SS_{\text{res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Simple Linear Regression

4. Divide and subtract to find R^2

Substitute your calculated sums into the main equation:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Simple Linear Regression

Practical Example

Imagine you have three data points: $(1, 2)$, $(2, 3)$, and $(3, 5)$.
Suppose your calculated linear regression line gives the predicted values $\hat{y} = [1.83, 3.33, 4.83]$.

x	Actual y	Predicted $\hat{y}$	$(y - \bar{y})^2$	$(y - \hat{y})^2$
1	2	1.83	$(2 - 3.33)^2 = 1.77$	$(2 - 1.83)^2 = 0.029$
2	3	3.33	$(3 - 3.33)^2 = 0.11$	$(3 - 3.33)^2 = 0.109$
3	5	4.83	$(5 - 3.33)^2 = 2.79$	$(5 - 4.83)^2 = 0.029$
Sum			$\boldsymbol{SS_{tot} = 4.67}$	$\boldsymbol{SS_{res} = 0.167}$

Using the formula:

$$R^2 = 1 - \frac{0.167}{4.67} = 1 - 0.0358 = 0.9642$$

- This means 96.42 of the variance in y is explained by the independent variable x

Multi Linear Regression

When we have more than one input features, Multi Linear Regression comes into picture, which is mostly the case in real world. In simple linear regression we try to fit the optimal line, but in multi we try to fit the plane(3D) or hyperplane (nD).

The formula for multiple linear regression is expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon = X\beta + \varepsilon$$

- y is the predicted value of the dependent variable
- β_0 is the y-intercept
- β_1 and β_p are the regression coefficients(or weights) for each independent variable (x_1 to x_p)
- ε represents the error term

Multi Linear Regression

In multiple linear regression, the β coefficients can be calculated using the following formula:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Where:

- X is the design matrix that includes all the independent variables, with a column of ones added for the intercept.
- y is the vector of observed values of the dependent variable.
- X^T is the transpose of the design matrix X.
- $(X^T X)^{-1}$ is the inverse of the matrix product $X^T X$.

Multi Linear Regression

Calculation Steps

1. **Construct the Design Matrix (X)**: Include all independent variables and a column for the intercept.
2. **Compute Transpose (X^T)**: Calculate the transpose of the design matrix.
3. **Calculate Matrix Product**: Find the product of X^T and X.
4. **Inverse Calculation**: Compute the inverse of the product from the previous step.
5. **Multiply with X^TY**: Multiply the inverse from step 4 with X^T and then with the dependent variable vector Y.

This process yields the estimated coefficients β, which describe the relationship between the independent variables and the dependent variable in the model.

Multi Linear Regression

Assume we have the following training data:

x_1	x_2	y
1	2	5
2	1	6
3	4	10
4	3	11

We want to find

- θ_0
- θ_1
- θ_2

- Notice that θ_1and θ_2 are obtained together, not separately.

θ $= \begin{bmatrix} 2.1 \\ 1.4 \\ 0.8 \end{bmatrix}$

Therefore,

$$\boxed{y = 2.1 + 1.4x_1 + 0.8x_2}$$